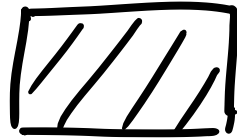


① P/L



$$\underline{CP = 10 \text{ ₹}}$$

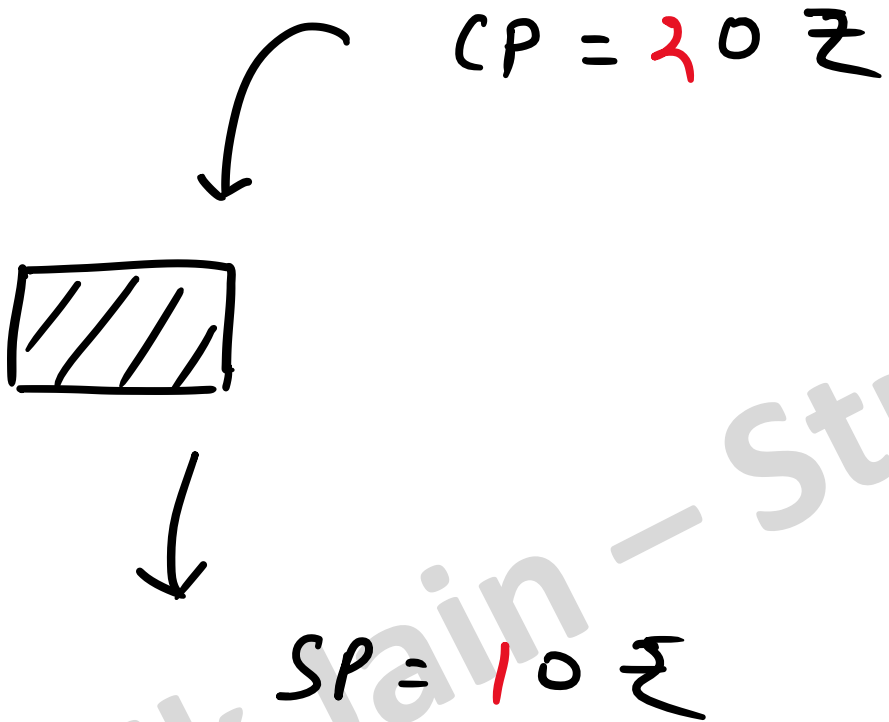


$$\underline{SP = 20 \text{ ₹}}$$

$$\Rightarrow \text{Profit} = ₹ 10$$

$$\Rightarrow \boxed{\text{Profit} = SP - CP} \rightarrow (+)$$

$$SP > CP$$



$$\Rightarrow \text{Loss} = ₹10$$

$$\Rightarrow \boxed{\text{Loss} = CP - SP} \rightarrow (+)$$

$CP > SP$

② % P, % L

↓

$$\% \text{ change} = \frac{FV - IV}{IV} \times 100$$

$$\% \text{ Profit} = \frac{SP - CP}{CP} \times 100$$

[% Gain]

$$\% \text{ Loss} = \frac{CP - SP}{CP} \times 100$$

③

$$\% \text{ Profit} = \frac{SP - CP}{CP} \times 100$$

$$\% P (CP) = (SP - CP) 100$$

$$\% P (CP) + CP (100) = SP \times 100$$

$$CP [\% P + 100] = SP \times 100$$

$\therefore$

$$SP = CP \left[1 + \frac{\% P}{100} \right]$$

$$\% \text{ Loss} = \frac{CP - SP}{CP} \times 100$$

$$\Rightarrow SP = CP \left[1 - \frac{\%L}{100} \right]$$

Q] Seema sells her watch for Rs 200 for a profit of 20% .At what price did Seema purchased it?

(M1)

$$\textcircled{1} \quad SP = ₹ 200$$

$$\%P = 20\%$$

$$CP = ?$$

$$\textcircled{2} \quad \%P = \frac{SP - CP}{CP} \times 100$$

$$20 = \frac{200 - x}{x} \times 100$$

$$x = ₹ 166.66 //$$

Q] Seema sells her watch for Rs 200 for a profit of 20% .At what price did Seema purchased it? ✓

①

$$SP = CP \times \frac{P}{100}$$

142

$$SP = CP \left[1 + \frac{\%P}{100} \right]$$

$$200 = x \left[1 + \frac{20}{100} \right]$$

$$x = ₹ 166.66 //$$

Q] Two lots of onion with equal quantity , one costing Rs 10 per kg and other costing Rs 15 per kg are mixed together and a lot is sold at Rs 15 per kg. What is Profit/Loss percentage?

$$\textcircled{i} \quad CP = ₹ 25 \quad [2\text{kg}]$$

$$\textcircled{M1} \quad SP = ₹ 30 \quad [2\text{kg}]$$

$$\begin{aligned} \% P &= \frac{SP - CP}{CP} \times 100 = \frac{30 - 25}{25} \times 100 \\ &= 20\% \end{aligned}$$

Q] Two lots of onion with equal quantity , one costing Rs 10 per kg and other costing Rs 15 per kg are mixed together and a lot is sold at Rs 15 per kg. What is Profit/Loss percentage?

$$\textcircled{1} \quad CP = ₹ 25 \quad [2\text{kg}]$$

$\textcircled{M2}$

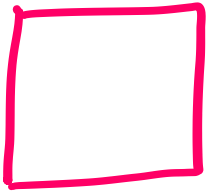
$$SP = ₹ 30 \quad [2\text{kg}]$$

$$SP = CP \left[1 + \frac{\%P}{100} \right]$$

$$30 = 25 \left[1 + \frac{\%P}{100} \right] \Rightarrow \%P = 20\% //$$

Ex:

$$\text{MRP} = ₹10$$



$$\text{SP} = ₹8$$

$$\text{Discount} = ₹2$$

$$\%D = \frac{10 - 8}{10} \times 100$$

$$= \frac{2}{10} \times 100 = 20\%$$

✓ 1] $\text{Discount} = \text{MP} - \text{SP}$

✓ 2] $\% \text{ Discount} = \frac{\text{MP} - \text{SP}}{\text{MP}} \times 100$

Shrenik Jain — Study Simplified

Q] On a 20% discount sale, an article costs Rs 596.
What was the original price of the article?

• Solⁿ : ① $\%D = 20\%$

$SP = ₹ 596$

$MP = ?$

mi

②

$$\%D = \frac{MP - SP}{MP} \times 100$$

(m2)

$$MP > \underline{SP}$$

$$SP = MP \times f$$

$$\underline{SP} = MP \left[1 - \frac{\%D}{100} \right]$$

$$596 = MP \left[1 - \frac{20}{510} \right]$$

$$MP = ₹ 745 //$$

Q] A person bought 8 quintal of rice for certain Rs.
After a week, he sold 3 quintal of rice at 10%
profit, 3 quintal at no Profit/loss and 2 quintal at
5% loss. In this transaction what is the profit%?

① $1\text{Q} \rightarrow ₹x$

$8\text{Q} \rightarrow ₹8x$

[Assume]

$CP = 8x$

② $SP_T = SP_1 + SP_L + SP_3 = 8.2x$

③ $\%P = \frac{8.2x - 8x}{8x} \times 100 = 2.5\% //$

(i) 30, 10% P

$$SP_1 = CP_1 \left[1 + \frac{10}{100} \right]$$

$$= 3x \left[1 + \frac{10}{100} \right] = 3.3x$$

(ii) 30, 0% P/L

$$SP_2 = CP = 3x$$

(iii) 24, 5% L

$$SP_3 = CP \left[1 - \frac{S}{100} \right]$$

$$= 2x \left[1 - \frac{5}{100} \right]$$

$$SP_3 = 1.9x$$

Q] A fruit seller sold a basket of fruits at 12.5% loss.
Had he sold it for Rs 108 more, he would have made a
10% gain. What is the loss in Rs incurred by the fruit
seller?

~~(A) 48~~

~~(B) 52~~

(C) 60

~~(D) 108~~

[GATE 2018 CE]

① [Assume] $CP = x = 480$

② $SP = \cancel{CP}^x \left[1 - \frac{12.5}{100} \right]$

$SP = \underline{0.875x} = 420$

③ $(0.875x + 108) = \cancel{CP}^x \left[1 + \frac{10}{100} \right]$

$x = 480$

$$\begin{aligned}\text{Loss} &= \text{CP} - \text{SP} \\ &= 480 - 420 \\ &= 60//\end{aligned}$$

Shrenik Jain — Study Simplified

Q] A firm is selling its product at Rs 60 per unit. The total cost of production is Rs 100 and firm is earning total profit of Rs 500. Later the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level?

$$SP - CP = P$$
$$SP = P + CP$$

- ~~(A) 5~~
- ~~(B) 10~~
- ~~(C) 15~~
- ~~(D) 30~~

[GATE 2013 CE]

$$\frac{600 \text{ ₹}}{60 \text{ ₹/unit}} = 10 \text{ units}$$

①

$$CP_T = 100$$

$$P_T = 500$$

$$SP_T = ?$$

$$\begin{aligned} SP_T &= P_T + CP_T \\ &= 500 + 100 \\ &= 600 // \end{aligned}$$

$$\textcircled{2} \quad CP_T' = \overset{100}{\cancel{CP_T}} \left[1 + \frac{30}{100} \right]$$

$$CP_T' = 130$$

$$P_T' = P_T = 500$$

$$SP_T' = P_T' + CP_T'$$

$$= 500 + 130$$

$$\underline{SP_T'} = 630$$

$$\Rightarrow SP' = \frac{630}{10} = 63 \text{ ₹}$$

$$\% \uparrow \text{ in SP} = \frac{FV - IV}{IV} \times 100$$

$$= \frac{SP' - SP}{SP} \times 100$$

$$= \frac{63 - 60}{60} \times 100$$

$$= 5\% //$$

Q] A foundry has a fixed daily cost of Rs 50,000 whenever it operates and a variable cost of Rs 800 Q, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes? [GATE 2014 XE/CH]

$$\textcircled{1} \quad C_T = \text{Fixed Cost} + \text{Variable cost}$$

$$= 50,000 + 800 \times 100$$

$$= 50k + 80k$$

$$C_T = 1,30,000 \text{ ₹} \longrightarrow 100 \text{ Tonnes}$$

$$\frac{1,30,000 \text{ ₹}}{100} \longrightarrow \frac{100 \text{ Tonnes}}{100}$$

$$1 \text{ Tonne} \Rightarrow 1,300 \text{ ₹}$$

Q] A refrigerator and a camera were sold for Rs 12000 each. The refrigerator was sold at a loss of 20% of the cost and the camera at a gain of 20% of the cost. The entire transaction results in which one of the following?

- (A) No loss or gain
- ☒ (B) Loss of Rs 1000
- (C) Gain of Rs 1000
- (D) Loss of Rs 2000

①

$$SP_R = 12K$$

$$SP_C = 12K$$

②

$$SP_R = CP_R \left[1 - \frac{20}{100} \right]$$

12K

$$CP_R = 15K //$$

③

$$SP_C = CP_C \left[1 + \frac{20}{100} \right]$$

12K

$$CP_C = 10K //$$

$$\therefore \left. \begin{array}{l} SP_Y = 12K \\ \underline{SP_C = 12K} \end{array} \right\} SP = 24K$$

$$\left. \begin{array}{l} CP_Y = 15K // \\ CP_C = 10K // \end{array} \right\} CP = 25K$$

$$SP < CP$$

$$\begin{aligned} \text{Loss} &= CP - SP = 25 - 24 \\ &= 1K // \end{aligned}$$

Q] A trader sells two cycles at Rs 1188 each and gains 10% on the first and lose 10% on the second. What is the profit or loss percent on the whole?

- ☒ (A) No loss or gain
(B) 1% gain
(C) 1% loss
(D) 2% loss

③ $\cancel{SP_2} = CP_2 \left[1 - \frac{10}{100} \right]$
1188

$CP_2 = 1320 //$

① $SP_1 = 1188$
 $SP_2 = 1188$

② $\cancel{SP_1} = CP_1 \left[1 + \frac{10}{100} \right]$
1188

$CP_1 = 1080 //$

$$SP = SP_1 + SP_2$$

$$= 2 \times 1188$$

$$= 2376$$

$$CP = CP_1 + CP_2$$

$$= \underline{\underline{2400}}$$

Loss

$$= CP - SP$$

$$= 2400 - 2376$$

$$= 24$$

$$\%L = \frac{24}{2400} \times 100$$

$$\boxed{\%L = 1\%}$$

